Problem-Solution Fit

Date	15 March 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

Validation of alignment between identified forensic investigative pain points and CrimeVision AI system capabilities.

Investigative Problem	Traditional Limitation	CrimeVision AI Solution	Validation & Fit Metric
Fragmented Evidence: Photos, audio, and video analyzed separately.	Manual, disconnected spreadsheets; high risk of contradictory conclusions.	Multimodal Fusion Service linking audio transcripts, CCTV timeline, and 2D bounding boxes.	**100% Fit:** Unified timeline aligning audio facts with CCTV visual events.
Flat 2D Photos: Lack depth and room perspective.	Inability to verify line-of-sight, scale, and exact distances between objects.	Three.js 3D Scene Reconstruction projecting depth coordinates [X,Y,Z] onto an explorable canvas.	**95% Fit:** Interactive 3D spatial walkthrough with camera reset and orbit controls.
Language Barriers: Non-English witness statements delayed.	Waiting days for accredited human translators stalls immediate leads.	Whisper ASR + Groq multilingual translation across English, Hindi, Telugu, and French.	**98% Fit:** Instant translation rendered in dedicated UI tabs in < 2 seconds.
Delayed Reporting: Manual court dossier preparation.	Hours of drafting susceptible to formatting flaws and evidentiary challenges.	One-click ReportLab PDF generation with metadata, evidence catalogs, and SHA-256 verification.	**100% Fit:** Exportable courtroom-standard forensic PDF dossier in under 1 second.